SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA0605 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning ***(BL 5)***

Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks:25**

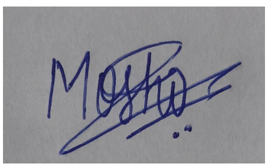
Annexure A

Industry Problem Solving

Code of Conduct:

I Moshika M - 192424064 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

Short Answer Test

1. A retail company wants to recommend products to customers based on past purchases. Design a solution using K-Nearest Neighbour or Case-Based Learning. Explain how similarity is measured, how recommendations are generated, and how the system adapts to new customer data.

ANSWER:

CODE

```python
import math
data = [
    ([5, 1, 2], "Electronics"),
    ([4, 2, 1], "Electronics"),
    ([1, 5, 4], "Clothing"),
    ([2, 4, 5], "Clothing"),
    ([5, 2, 1], "Electronics")
]
k = int(input("Enter K value: "))
books = int(input("Enter number of books purchased: "))
electronics = int(input("Enter number of electronics purchased: "))
clothing = int(input("Enter number of clothing items purchased: "))
test = [books, electronics, clothing]
distances = []
for features, label in data:
    d = math.sqrt(sum((features[i] - test[i]) ** 2 for i in range(3)))
    distances.append((d, label))
distances.sort()
neighbors = distances[:k]
count = {}
for d, label in neighbors:
    count[label] = count.get(label, 0) + 1
prediction = max(count, key=count.get)
print("Recommended Product Category:", prediction)
```

OUTPUT

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/Industry Problem Solving.py
Enter K value: 3
Enter number of books purchased: 5
Enter number of electronics purchased: 1
Enter number of clothing items purchased: 2
Recommended Product Category: Electronics

Process finished with exit code 0
```

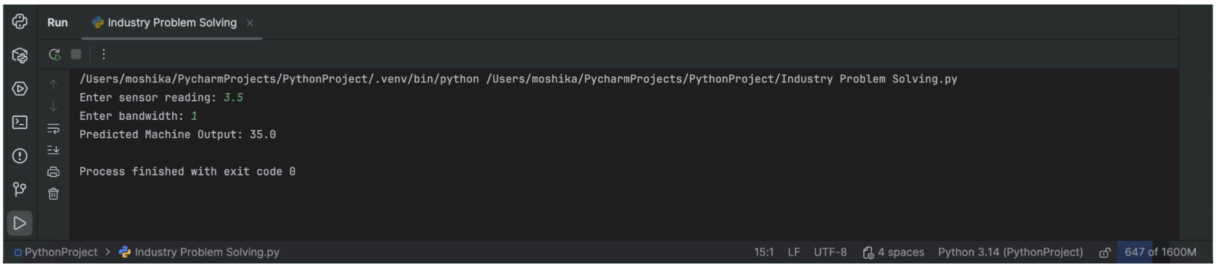
2. A manufacturing industry needs to predict machine output based on sensor readings that vary over time. Propose a solution using Locally Weighted Regression. Explain how local models improve prediction accuracy and how the system handles noisy or incomplete data.

ANSWER:

CODE

```python
import numpy as np

x = np.array([1, 2, 3, 4, 5])
y = np.array([10, 20, 30, 40, 50])

test = float(input("Enter sensor reading: "))
tau = float(input("Enter bandwidth: "))

weights = np.exp(-((x - test) ** 2) / (2 * tau ** 2))

X = np.vstack([np.ones(len(x)), x]).T

W = np.diag(weights)

theta = np.linalg.inv(X.T @ W @ X) @ X.T @ W @ y

prediction = theta[0] + theta[1] * test

print("Predicted Machine Output:", round(prediction, 2))
```

OUTPUT

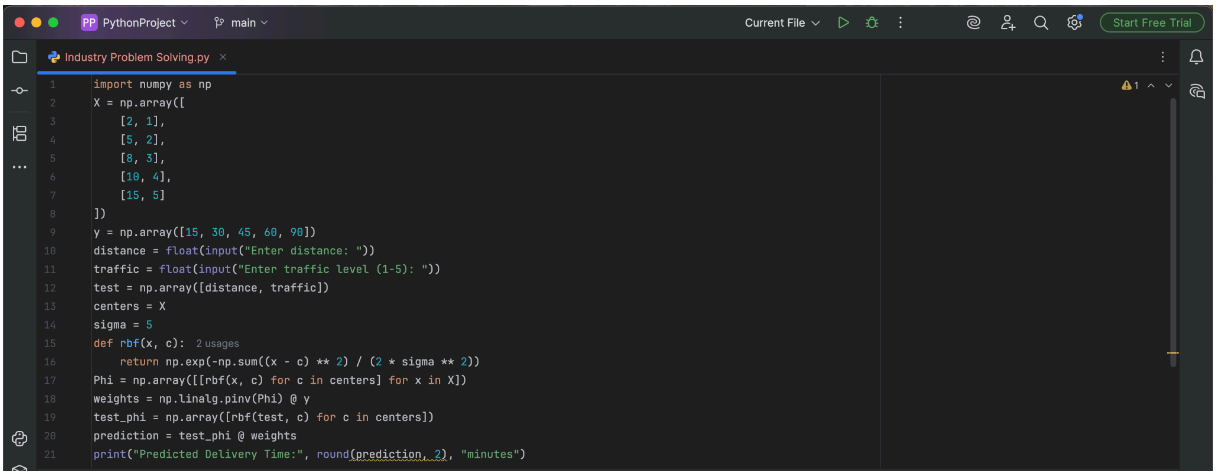


```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/Industry Problem Solving.py
Enter sensor reading: 3.5
Enter bandwidth: 1
Predicted Machine Output: 35.0

Process finished with exit code 0
```

5. A logistics company needs to forecast delivery times based on location and traffic patterns. Propose a solution using Radial Basis Function networks or Locally Weighted Regression. Explain how the model captures non-linear relationships and improves prediction reliability under varying conditions.

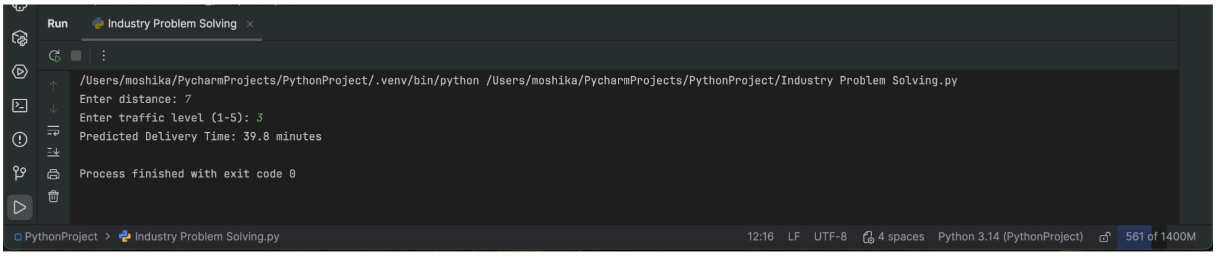
ANSWER:

CODE



```python
import numpy as np
X = np.array([
    [2, 1],
    [5, 2],
    [8, 3],
    [10, 4],
    [15, 5]
])
y = np.array([15, 30, 45, 60, 90])
distance = float(input("Enter distance: "))
traffic = float(input("Enter traffic level (1-5): "))
test = np.array([distance, traffic])
centers = X
sigma = 5
def rbf(x, c):
    return np.exp(-np.sum((x - c) ** 2) / (2 * sigma ** 2))
Phi = np.array([[rbf(x, c) for c in centers] for x in X])
weights = np.linalg.pinv(Phi) @ y
test_phi = np.array([rbf(test, c) for c in centers])
prediction = test_phi @ weights
print("Predicted Delivery Time:", round(prediction, 2), "minutes")
```

OUTPUT



```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/Industry Problem Solving.py
Enter distance: 7
Enter traffic level (1-5): 3
Predicted Delivery Time: 39.8 minutes

Process finished with exit code 0
```

6. A banking system wants to detect potential loan defaulters based on customer financial history and transaction patterns. Design a solution using K-Nearest Neighbour or Logistic Regression. Explain how similarity or decision boundaries are used, how features are selected, and how the system adapts to new customer data over time.

ANSWER:

CODE

```python
import numpy as np
X = np.array([
    [20, 500],
    [25, 520],
    [30, 550],
    [40, 600],
    [50, 650],
    [60, 700],
    [70, 750],
    [80, 800]
])
y = np.array([0, 0, 0, 0, 1, 1, 1, 1])
X_mean = X.mean(axis=0)
X_std = X.std(axis=0)
X = (X - X_mean) / X_std
X = np.c_[np.ones(len(X)), X]
weights = np.zeros(X.shape[1])
def sigmoid(z):
    return 1 / (1 + np.exp(-z))
for i in range(1000):
    prediction = sigmoid(X @ weights)
    weights += 0.1 * X.T @ (y - prediction)
income = float(input("Enter customer income: "))
credit = float(input("Enter credit score: "))
test = np.array([
    1,
    (income - X_mean[0]) / X_std[0],
    (credit - X_mean[1]) / X_std[1]
])
probability = sigmoid(test @ weights)
if probability >= 0.5:
    print("Prediction: No Default")
else:
    print("Prediction: Default")
print("Probability:", round(probability, 2))
```

OUTPUT

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/Industry Problem Solving.py
Enter customer income: 65
Enter credit score: 720
Prediction: No Default
Probability: 1.0

Process finished with exit code 0
```

8. A cybersecurity firm wants to identify abnormal network behavior to detect intrusions. Design a system using KNN or Case-Based Learning. Explain how similarity between network patterns is measured, how anomalies are detected, and how the model updates with new attack patterns.

ANSWER:

CODE

```python
import math
data = [
    ([10, 2], "Normal"),
    ([12, 3], "Normal"),
    ([15, 3], "Normal"),
    ([50, 20], "Attack"),
    ([60, 25], "Attack"),
    ([55, 22], "Attack")
]
k = int(input("Enter K value: "))
packets = int(input("Enter number of packets: "))
connections = int(input("Enter number of connections: "))
test = [packets, connections]
distances = []
for features, label in data:
    d = math.sqrt(
        (features[0] - test[0]) ** 2 +
        (features[1] - test[1]) ** 2
    )
    distances.append((d, label))
distances.sort()
neighbors = distances[:k]
normal = sum(1 for d, label in neighbors if label == "Normal")
attack = sum(1 for d, label in neighbors if label == "Attack")
if attack > normal:
    result = "Attack Detected"
else:
    result = "Normal Network Behavior"
print("Result:", result)
```

OUTPUT

```
/Users/moshika/PycharmProjects/PythonProject/.venv/bin/python /Users/moshika/PycharmProjects/PythonProject/Industry Problem Solving.py
Enter K value: 3
Enter number of packets: 52
Enter number of connections: 21
Result: Attack Detected

Process finished with exit code 0
```

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**25**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand